Introductory To Programming and Problem-Solving Skills

Tutorial Week -4

Control Structures

Basanta Singh

Let’s Play a Game!!

- Let’s make a group of 5-6 students (Total: 5 groups). Each group will be handed a ball, and the ball should be circulate within the group. But, in order to pass the ball, each student should answer a question from the past course - content.
- Note: A timer should be there in order to track down the activity as per the group. The group that ends with in the minimum time will win the game.

Repetition

What is Loop?

Loop is a sequence of instructions that is **continuously repeated** until a certain **condition is reached**. It is also said to be repetition control structure.

Do we repeat our work in real time?
More importantly, why do we need this concept in programming??

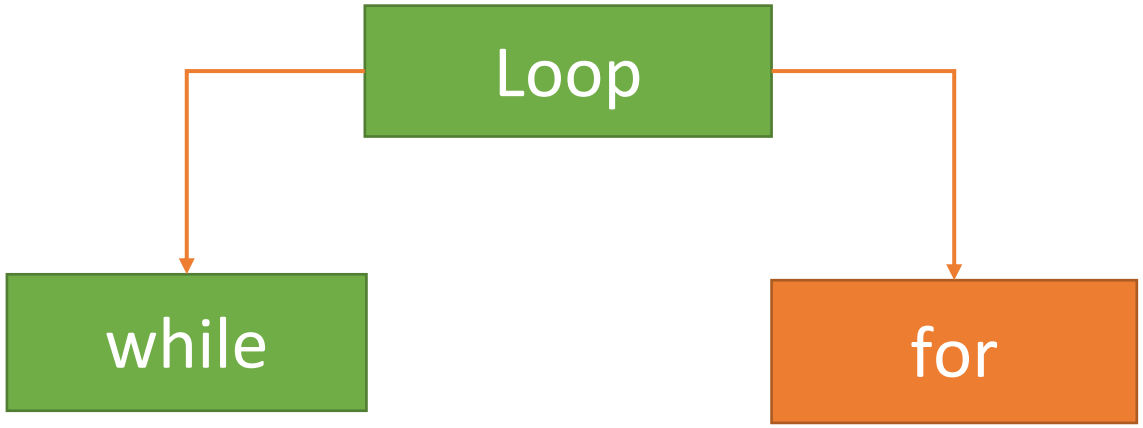
BEFORE GOING FURTHER!!

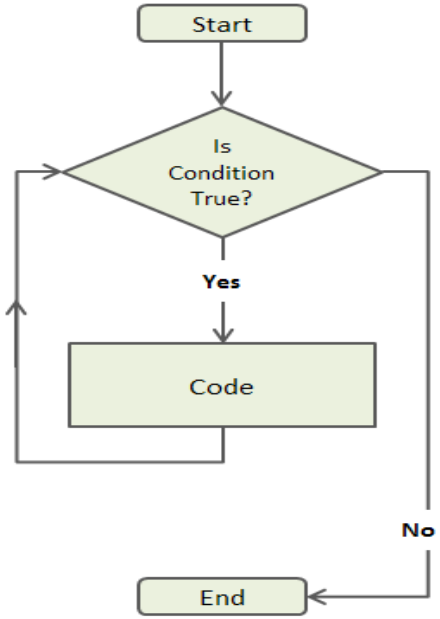
- How will you develop a program that prints from 1 to 5?
- Or even, how would you develop a program that prints from 1 to 1000000?

Here comes, the concept of loop!!

Types of Loop

Technically, in python there are two types of loop.

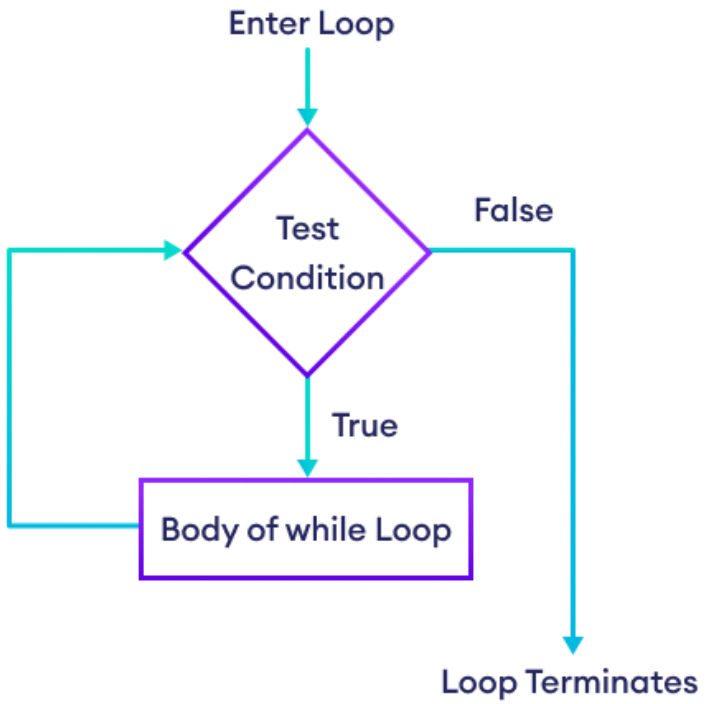




Both types of loops serve distinct purposes and have their own significance in programming. Understanding when and how to use each type is essential for effective implementation.

while Loop

- In python, ‘**while**’ loop is a control flow structure that allows a block of code to be executed **until the given condition is true**.



Syntax

```
while <<condition>>:
    # body of while loop
```

Indentation

Example 1

Program to display numbers from 1 to 5.

```
# program to display numbers from 1 to 5

# initialize the variable
i = 1
n = 5

# while loop from i = 1 to 5
while i <= n:
    print(i)
    i = i + 1
```

From where?

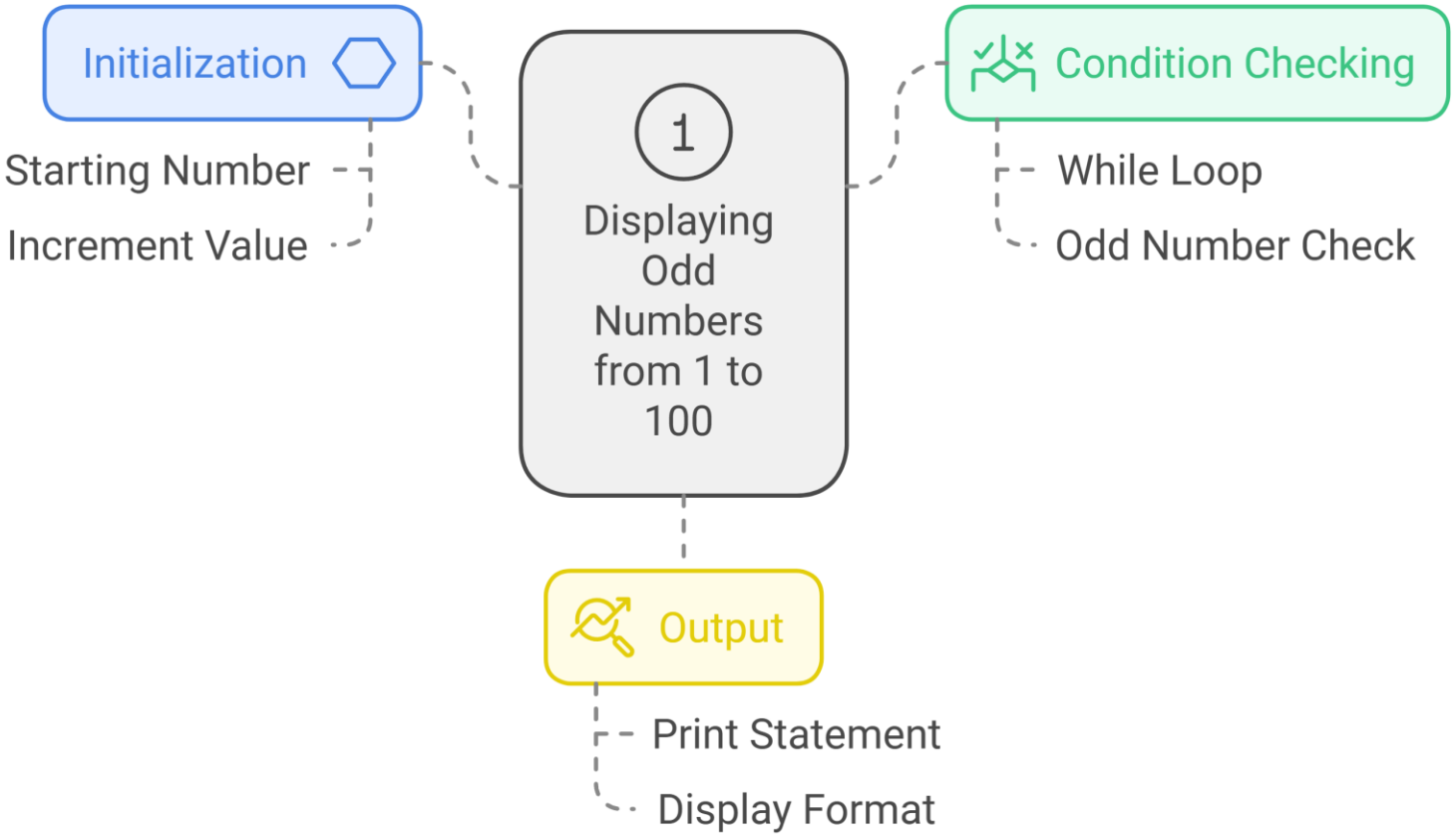
To what?

Condition?

Loop Terminate?

For You

1. Develop a program that iterates from 1 to 100. However, it displays only odd number



For You

1. Develop a program that iterates from 1 to 100. However, it displays only odd numbers.

```
# Program to display odd numbers from 1 to 100 using a while
loop
number = 1  # Start from 1
while number <= 100:  # Continue until the number reaches 100
    if number % 2 != 0:  # Check if the number is odd
        print(number)  # Display the odd number
    number += 1  # Increment the number by 1
```

while Loop – Input Error Checking

- In python, we can use while loop to check user's input and allow user to re-enter if there is no desired input.

```
valid_input = False
while not valid_input:
        age = int(input("Enter your age: "))

        if age < 0:
            print("Invalid input. Age cannot be negative.")
        else:
            valid_input = True

print(f"Your age is {age}.")
```

```
Enter your age: -1
Invalid input. Age cannot be negative.
Enter your age: -2
Invalid input. Age cannot be negative.
Enter your age: 0
Your age is 0.
```

For You

Develop a menu dashboard, where user will have three choices.

1. Print Paycheck
2. Change Benefits
3. Exit

Validate, whether the user is choosing apart from those 3 option!!

```
while True:  # Infinite loop to keep showing the menu until Exit is chosen
    print("\n--- Menu Dashboard ---")
    print("1. Print Paycheck")
    print("2. Change Benefits")
    print("3. Exit")

    choice = input("Enter your choice (1-3): ")  # Take user input as a string

    if choice == "1":
        print("\n[Paycheck]: Your paycheck has been printed.")
    elif choice == "2":
        print("\n[Benefits]: You can now change your benefits.")
    elif choice == "3":
        print("\n[Exit]: Thank you! Goodbye!")
        break  # Exit the loop to terminate the program
    else:
        print("\n[Error]: Invalid choice. Please select 1, 2, or 3.")  # Validation
for invalid choice
```

while Loop – Sentinel Values

- How would you handle when you need to sum up the input from the user? Up to what extent should the input be accumulated, and what criteria would determine when the process should stop?

THINK ABOUT IT!!

- In computer programming, a sentinel value is a value or input in the context of an algorithm which signifies that the program should terminate.

Example 3

Develop a program that prompts user to enter a series of numbers and sum them until they enter an empty string.

```
sum_num = 0
user_input = input("Enter a number: ")
while user_input != "":
    sum_num = sum_num + int(user_input)
    user_input = input("Enter a number: ")
print(f"Total Number is {sum_num}")
```

```
Enter a number: 45
Enter a number: 45
Enter a number:
Total Number is 90
```

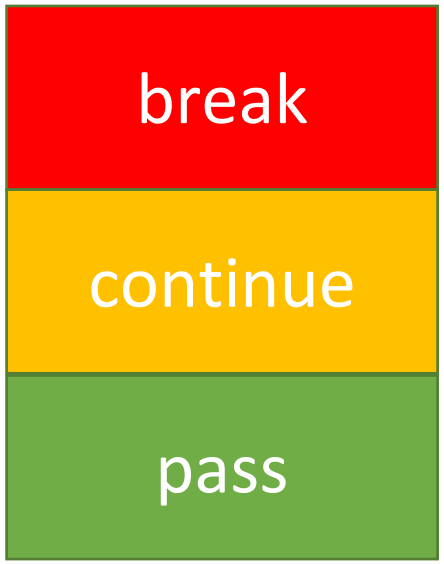
Here, blank string is the sentinel value.

For You

Write a python program that asks the user to enter a series of positive integers. The program should keep prompting the user for input until they enter a negative number.

Additional Iteration Statement

- In Python, loops iterate over a block of condition until the given condition is false. However, there are times where we want to terminate our loop before hand. There are several additional iteration to do so.



break statement

- Break statement is used to terminate the loop immediately when it is encountered.

  The syntax of the break statement is: **break**

```
# program to find first 5 multiples of 6

i = 1

while i <= 10:
    print('6 * ',(i), '=',6 * i)

    if i >= 5:
        break

    i = i + 1
```

The program prints the first 5 multiples of 6 but stops printing them once it reaches the 5th multiple.

```
6 *  1 = 6
6 *  2 = 12
6 *  3 = 18
6 *  4 = 24
6 *  5 = 30
```

continue statement

- Continue statement is used to skip the current iteration of the loop. Then the control of the programs jumps to the next iteration. The syntax of the continue statement is:

continue

```
i = 1

while i <= 10:
    if i == 5:
        i += 1
        continue
    mul = 6 * i
    print(f"6 * {i} = {mul}")
    i += 1
```

```
6 * 1 = 6
6 * 2 = 12
6 * 3 = 18
6 * 4 = 24
6 * 6 = 36
6 * 7 = 42
6 * 8 = 48
6 * 9 = 54
6 * 10 = 60
```

The program prints the first 4 multiples of 6 but stops printing them once it reaches the 5th multiple.
And continue again from 6th multiple.

Example 6

break vs. continue

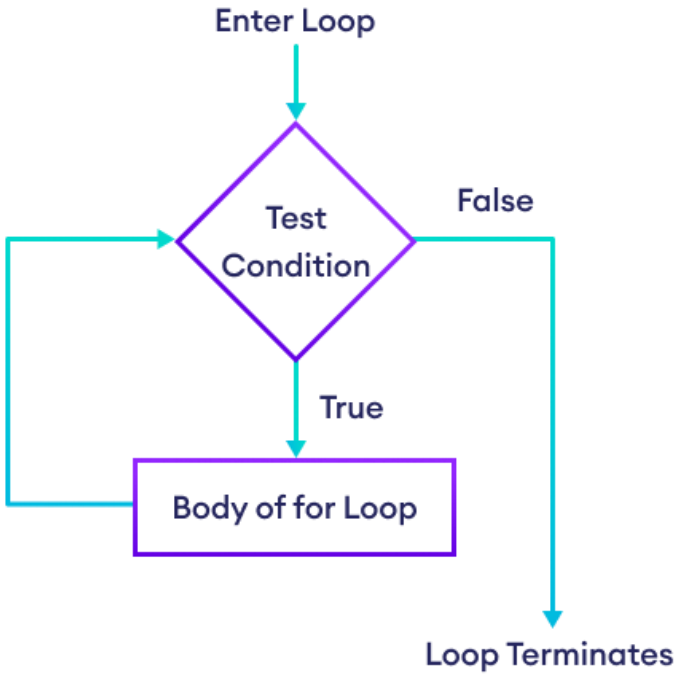
```python
sum = 0
while True:
    num = int(input("Enter a positive integer (or enter 0 to stop): "))
    if num == 0:
        break
    elif num < 0:
        print("Invalid input. Please enter a positive integer.")
        continue
    elif num % 2 == 0:
        print("Skipping even number.")
        continue
    else:
        sum += num
        print(f"The current sum is {sum}.")

print(f"The final sum is {sum}.")
```

```
Enter a positive integer (or enter 0 to stop): 1
The current sum is 1.
Enter a positive integer (or enter 0 to stop): 2
Skipping even number.
Enter a positive integer (or enter 0 to stop): 4
Skipping even number.
Enter a positive integer (or enter 0 to stop): 0
The final sum is 1.
```

for Loop

- In python, '**for**' loop is a control flow structure that allows you to iterate over a sequence of elements such as string, list, tuple and so on.



Indentation

Syntax

```
for variable in sequence:
    # do something with variable
```

Example 7

Program to display numbers from 1 to 5.

```
numbers = [1,2,3,4,5]
for i in numbers:
    print(i)
```

From where?

Sequence!!

Loop Variable.

Loop Terminate?

Sequence could be generated with the help of range function.

for Loop – range()

- In python, '**range()**' is a built in function that creates a sequence of numbers that can be used for various purposes, such as looping over a block of code a certain number of times.

```
for i in range(1,11):
    print(i)
```

```
for i in range(1,11,2):
    print(i)
```

Syntax

range(start, stop, step)

Remember: The range function in Python is inclusive of the starting value and exclusive of the ending value.

Nested Loop

What is Nested Loop?

Nested loops in python refer to the situation where one loop is placed inside another loop.

Remember: During the nested loop, there is the concept of outer loop and inner loop. We will discuss this further.

Example 8

Program to display a half pyramid pattern in star symbol.

```
#initialisation of outer loop
for i in range(1,6):
    #initialisation of outer loop
    for j in range(i):
        #displays the number
        print("*", end = " ")
        #new line after each itteration of inner loop
    print()
```

```
*
* *
* * *
* * * *
* * * * *
```

Outer Loop

end = “ ”

Range

Inner Loop

For You

Program to display a downward half pyramid pattern in star symbol.

For You – Exercise

1. Write a program to print multiplication of a user's input number till 10.
2. Write a program that iterates the integer from 1 to 50. For multiple of three print "Fizz" instead of the number and for multiples of five, print "Buzz". For the number of 3 and 5 print "FizzBuzz"
3. Write a python program to enter the numbers till the user wants and at the end it should display the count of positive number, negative number and zeros entered.
4. Develop a program to develop a pattern of full pyramid in star.

For You – Additional Questions

1. Write a Python program for a number guessing game where the computer chooses a random number between 1 and 20, and the player has five chances to guess the number. Use a for loop to limit the number of guesses the player can make. After each guess, the program should tell the player whether the guess was too high or too low. If the player correctly guesses the number, the program should congratulate the player and end the game. If the player runs out of guesses, the program should reveal the number and show the player's guesses.

You Should Include The Concept of For Loop, Formatted Output, import random module. Comments are also mandatory.